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Effects of dietary cation-anion difference and nicotinic acid supplementation on prepartum Jersey cows and calves: II—Effects on calf performance

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ABSTRACT

Two DCAD (−40 or −80 mEq/kg) $\pm$ 23 g/d supplemental nicotinic acid (NA) were fed to prepartum Jersey cows. The effects of NA and DCAD supplementation of prepartum Jersey cows on the growth and performance of their calves have not been studied. Briefly, 40 multiparous Jersey cows were blocked by calving date and assigned to treatments 4 wk prepartum in a 2 $\times$ 2 factorial arrangement of treatments in a randomized complete block design. Colostrum was collected within 90 min after parturition, and calves were not allowed to suckle. Calves were weighed and had blood sampled within 30 min of birth and were enrolled if their dams produced $\geq$ 2 L colostrum. Enrolled calves received $\geq$ 2 L and a maximum of 3.78 L. However, there were no differences in colostrum intake. The calves were blocked based on treatments of dams and fed twice daily from d 1 to 42: 3.78 L milk (13.3% to 8.5% BW from birth to 6 wk), pelleted calf starter with additional steam-flaked corn, and ad libitum water. A lower colostral IgG intake was observed in calves from −80 DCAD cows as compared with calves from −40 DCAD cows. An interaction was found for IGF-1 intake in which the greatest intake was observed in calves from cows fed −40/NA⁺. Colostral lactoferrin intake was greater in calves from cows fed NA. Xylose challenge data supported this, where calves from NA cows had greater plasma xylose. No differences were observed for calf plasma 24 h IgG, apparent efficiency of absorption, starter, milk solids, and water intakes, total DMI, weekly BW, final BW, skeletal measurements (overall and daily), weekly and final plasma glucose, and serum ketones. There was an interaction for calf final ketones, where the lowest were from −40/

NA[−] and −80/NA⁺. Calves from cows fed the −80 DCAD and cows fed NA resulted in a tendency to decrease calf ADG. Feed efficiency results indicated that −80 DCAD and NA decreased calf feed efficiency. These data suggest that supplementing NA in a −80 DCAD diet is not recommended, and providing prepartum Jersey cows with a lesser NA dose may be beneficial due to its effects on calf xylose absorption and lactoferrin intake.

Key words: colostrum, feed efficiency, lactoferrin, xylose absorption, insulin-like growth factor-1

INTRODUCTION

Gestational nutrition and metabolic status of the dam can affect the calf (van Niekerk et al., 2021; Mee, 2023) through fetal programming and colostrum intake, which ultimately impact calf vigor, neonatal immunity, and health (Mee, 2023). Therefore, important consideration needs to be taken regarding the health and physiology of the calf, specifically through the management of the dam (Davis and Drackley, 1998) and her colostrum production. It is well established that colostrum contains less lactose than mature milk and is comprised of higher amounts of fat, protein, vitamins and minerals, and non-nutrient substances, such as Ig, peptides, biologically important proteins (e.g., lactoferrin), hormones (e.g., prolactin and insulin), growth factors (e.g., IGF I and II, transforming growth factor- β (TGF β) I and II), cytokines, and nucleotides (Rauprich et al., 2000; van Niekerk et al., 2021; Erickson, 2022). The consumption of these important compounds in colostrum, as well as milk intake, is vital in their impacts on calf growth, gut development, and future milk production (van Niekerk et al., 2021).

Due to its importance to calf survival, it is vital to ensure that the dam produces enough good-quality colostrum (>50 g/L IgG; USDA NAHMS, 2021). However, it has been established that many multiparous Jersey cows have reduced colostrum production (Gavin et al., 2018; Stahl et al., 2024). A companion paper (Stahl et al., 2026) discussed the effects on cow performance when multiparous Jersey cows were fed 2 levels of DCAD diets with or without nicotinic acid (NA), hypothesizing that

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

altering prepartum DCAD levels would aid in improving colostrum quantity and that NA would enhance quality and increase bioactive compound concentration.

There is conflicting data on DCAD diets fed to the dam and their effects on the subsequent calf, with one study finding calves born to cows that were fed a negative DCAD (-115 mEq/kg DM) had lower preweaning BW and greater incidence of scours (Rajaeerad et al., 2020), but others saw no effect of negative DCAD level on calf performance (Collazos et al., 2017; Zimpel et al., 2021). Melendez et al. (2021) also observed that reducing the dam's urine pH to <6.0 through the utilization of anionic salts in a negative DCAD diet can significantly increase the risk of the dam having a stillbirth. However, data with sows showed that those with lower urine pH (pH <7.0) and greater serum Ca had fewer stillborn piglets (Craig et al., 2024; $P < 0.001$). For NA, it has been observed that prepartum supplementation in Holstein cows improved colostrum quality (IgG concentration; Aragona et al., 2016, 2020), as well as enhanced feed efficiency (FE) in their calves (Aragona et al., 2020). Aragona et al. (2020) hypothesized that the increase in colostrum IgG concentration was due to an increase in ruminal microbial protein synthesis, indicated by enhanced urinary purine derivatives. There is also research that indicated that when cows were supplemented with NA, there was an increase in vasodilation (Tunaru et al., 2003; Benyó et al., 2006). These effects are hypothesized to increase colostrogenesis, specifically increasing the amount of available AA for absorption and transport to the mammary gland, which could result in increased IgG production. Based on the hypothesized improvement of colostrogenesis, this may result in increased bioactive compound concentrations in colostrum. When considering Aragona et al. (2020) FE data, this improvement in calf performance could be due to fetal programming or alterations in bioactive compound concentrations in colostrum that would improve calf intestinal development. To our knowledge, there has been no research evaluating Jersey cow and calf performance when different DCAD concentrations are fed or through the feeding of NA.

The objectives of this study were to evaluate colostrum characteristics of cows that were fed 2 different DCAD levels with or without 23 g/d NA and its effect on their calves. The current study aimed to evaluate the effects of Jersey calf performance based on the treatment of their dams because there are data using Holstein cows where negative, prepartum DCAD diets (Collazos et al., 2017; Rajaeerad et al., 2020; Melendez et al., 2021; Zimpel et al., 2021) or NA supplementation (Aragona et al., 2016, 2020) can impact the calf. We hypothesized that altering prepartum DCAD levels would increase colostrum quantity to provide to their respective calves, NA would improve quality and increase select bioactive compound

concentrations that influence gastrointestinal tract development, and that those concentrations in NA-supplemented cow colostrum would improve calf intestinal development, resulting in greater calf growth and FE preweaning.

MATERIALS AND METHODS

Experimental Design and Animal Management

This experiment was reviewed and approved by the University of New Hampshire Institutional Animal Care and Use Committee (Protocol #200202). It was conducted at the University of New Hampshire Burley-DeMeritt Organic Dairy Research Farm in Lee, New Hampshire, from June 2020 to December 2021.

The effects of 2 levels of DCAD (-40 or -80 mEq/kg) $\pm$ 23 g/d unprotected NA fed to Jersey cows and the subsequent effects on their calves after being fed their dam's colostrum were studied in a 2×2 factorial arrangement of treatments in a randomized complete block design. The DCAD equation used was $[(\text{Na}^+ + \text{K}^+) - (\text{Cl}^- + \text{S}^{2-})]$ (Ender et al., 1971). Nicotinic acid dosage was estimated using 32 g/d (Aragona et al., 2020) and calculated based on the metabolic body size ($\text{BW}^{0.75}$) of Jerseys using the following data: Prepartum Jersey BW was assumed to be ~ 520 kg ($\text{BW}^{0.75} = 108.9$ kg), whereas the prepartum BW of the Holstein cows in the study of Aragona et al. (2020) was 801 kg ($\text{BW}^{0.75} = 150.6$ kg) for the cows fed 32 g/d NA. Therefore, based on metabolic body size, the NA dose was 23 g/d or ~ 211 mg/kg $\text{BW}^{0.75}$.

Forty multiparous Jersey cows were blocked by expected calving date and randomly assigned to 1 of 4 treatments: (1) -40 mEq/kg of DM DCAD without NA ($-40/\text{NA}^-$); (2) -40 mEq/kg of DM DCAD with 23 g/d NA ($-40/\text{NA}^+$); (3) -80 mEq/kg of DM DCAD without NA ($-80/\text{NA}^-$); and (4) -80 mEq/kg of DM DCAD with NA ($-80/\text{NA}^+$). Cow feeding and sampling are described in detail in a companion paper (Stahl et al., 2026). Briefly, cows were fed individually and housed on a compost-bedded pack. Treatments were administered beginning 4 wk prepartum and continued until calving. If cows did not calve before their expected calving date, they were moved into individual maternity pens 3 d before the expected calving date. Urine pH did not differ between the main effect of DCAD.

Parturition: Colostrum Collection and Feeding

All colostrum was collected within 60 to 90 min of calving. Calves were removed from dams within 30 min of birth and were not allowed to suckle. Calves were weighed and placed into individual pens (1×2.15 m) in a naturally ventilated enclosed calf barn. Navels were dipped in a 7% iodine tincture and received an oral vacci-

nation (Tri-Shield First Defense, ImmuCell Corporation, Portland, ME). Calves had to receive an amount of maternal colostrum between a minimum of 2 L and a maximum of 3.78 L to enroll in trial. This was decided before the trial, knowing that colostrum yields for multiparous Jersey cows are variable (Gavin et al., 2018), and was based on recommendations from the Bovine Alliance on Management and Nutrition (1995) of feeding only 1.89 L of good-quality colostrum at birth (Jaster, 2005). Colostrum was fed up to 2.5 L via nipple bottle within 2 h of birth, and then the remainder at 6 h. If calves refused to consume colostrum via nipple bottle, the colostrum was provided via esophageal feeder. If some colostrum was refused, it was kept warm and reintroduced 1 h later. Then, if the remaining colostrum was refused, it was fed via an esophageal feeder.

The enrolled calves born from study cows received treatments in utero via maternal blood supply and were, therefore, assigned the same treatments and blocks as their dams. Colostrum was analyzed for IgG, TGF- β 1, TGF- β 2, IGF-1, and LF as described in the companion paper (Stahl et al., 2026), and their intake was calculated.

Calf Feeding, Sampling, and Analyses

After 24 h, calves were fed 3.78 L/d of bulk-tank whole milk (WM; Table 1). Calves were fed 1.89 L twice daily at 0600 and 1800 h. Milk intake as a percent of birth BW was 13.2%, 13.2%, 13.6%, and 13.2% for -40/NA⁻, -40/NA⁺, -80/NA⁻, and -80/NA⁺, respectively. Throughout the entire study, bottles of WM would be filled in advance and frozen for future use. When bottles were filled, an aliquot was immediately collected and stored at 4°C for future analysis. The WM samples were shipped overnight (after collection) in ice packs to Dairy One Laboratories (Ithaca, NY) to be analyzed for fat, protein, solids, lactose, SCC, and MUN by Fourier Transform Infrared Spectroscopy using a Milkoscan FT+ (Foss, Inc., Hilleroed, Denmark). Calves were fed a 22.3% CP coarse starter (Table 1; Morrison's Custom Feeds, Inc., Barnet, VT) and water ad libitum starting at 1 d of age until 42 d. The starter was mixed and analyzed by the mill for nutrient content (Morrison's Custom Feeds, Inc., Barnet, VT; Table 1) and contained steam-flaked corn and a protein pellet (soy meal, wheat middlings, corn meal, barley, roasted soy, molasses, kelp meal, and vitamins and minerals). The amount of starter and water fed and refused was measured once daily at 0600 h. If amounts of starter and water were low at the second WM feeding at 1800 h, they were refilled and recorded.

The current experiment did not monitor health status or fecal scoring for statistical analysis. Health status was monitored via observation by researchers and farm staff. Rectal temperatures and observational fecal scoring were

conducted if calves displayed abnormal behavior or milk intake and were treated with organic medication per instructions by the station veterinarian.

Calf Skeletal and Blood Sampling and Analyses

At the time of parturition and before receiving maternal colostrum, a blood sample was collected via jugular venipuncture from 40 of the 41 calves (one calf was a stillborn) born within 30 min for determination of initial (0 h) serum IgG. These 40 calves also had birth weight recorded at 0 h for determination of initial BW. The enrolled calves then had a blood sample collected at 24 h for determination of serum IgG. Both the 0 and 24 h blood samples were collected via jugular venipuncture into a 10 mL tube with no additive (Monoject, Covidien, Mansfield, MA) and were allowed to clot before centrifuging at $1,278 \times g$ at 4°C for 20 min (5430R, Eppendorf, Hamburg, Germany). The serum was aspirated and stored at -20°C until further analysis of IgG via radial immunodiffusion assay (Triple J Farms, Bellingham, WA). Apparent efficiency of absorption (AEA) at 24 h of age was estimated using the following equation, according to Quigley et al. (1998):

$$[24 \text{ h plasma IgG (g/L)} \times \text{BW (kg)} \\ \times 0.09] / \text{IgG intake (g/L)} \times 100.$$

Additional samples were collected weekly for 42 d. Weekly sampling dates were based on calf birth dates as follows: calves born after 1630 h on Thursday until Monday at 0700 h were sampled on Tuesdays at 0700 h, and calves born on Monday after 0700 h until Thursday

Table 1. Nutrient analysis of calf starter and milk

Item	Starter ^{1,2}	Milk ³
CP	22.30	3.66 ± 0.14
ADF	7.57	—
NDF	18.31	—
NFC	48.76	—
Starch	34.61	—
Crude fat	5.56	4.88 ± 0.49
Ash	7.90	—
Lactose	—	4.73 ± 0.09
Solids	—	14.31 ± 0.92
SCC ⁴	—	138.13 ± 48.10
MUN ⁵	—	12.00 ± 1.78

¹Ingredient composition of starter (as fed): 21% organic soy meal, 20% organic wheat middlings, 17.3% organic corn meal, 12.4% organic barley, 10% organic steamed flake corn, 9.3% organic roasted soy, 6% organic molasses, 3.3% vitamins and minerals, and 0.7% kelp meal.

²Nutrient analysis of starter % DM basis.

³Nutrient analysis of milk as fed (mean ± SD).

⁴1,000 cells per mL.

⁵mg/dL.

at 1630 h were sampled on Fridays at 1630 h. During sampling, blood samples were collected, and BW measurements and skeletal measurements were taken. Plasma and serum blood samples were collected via jugular venipuncture into a 10 mL tube with EDTA (Monoject, Covidien, Mansfield, MA) and a 10 mL tube with no additive (Monoject, Covidien, Mansfield, MA), respectively. After collection, blood samples were placed on ice and transported to the laboratory. At the laboratory, a single drop of whole blood was taken from the 10-mL tube with no additive for immediate analysis of ketones using a NovaMax Plus Ketone meter (Nova Max Plus, Nova Biomedical, Waltham, MA). Ketone concentration was measured in duplicate. The tube with EDTA was centrifuged at $3,300 \times g$ at 4°C for 20 min (5430R, Eppendorf, Hamburg, Germany). Plasma was aspirated and stored at -20°C until further analysis for glucose. Plasma glucose concentrations were measured in duplicate using the Wako Autokit Glucose Assay (Wako Diagnostics, Mountain View, CA).

Body weight and skeletal measurements, including hip height, withers height, body length, and heart girth, were recorded weekly. Withers and hip heights were measured using a sliding-scale height stick with a bubble level. Body length and heart girth were measured using a weight tape (The Coburn Company, Inc., Whitewater, WI). Skeletal measurements were used to calculate gain in centimeters per day.

Calf Xylose Challenge

On d 5 of life, calves were subjected to a xylose challenge. D-Xylose is used as an indirect marker for glucose absorption across the small intestine (Hammon and Blum, 1997; Kuhne et al., 2000; Rauprich et al., 2000). D-xylose has been shown to have a low affinity for the Na^{+} -dependent glucose transporter in the bovine brush border membrane. Thus, it is more selectively absorbed and not metabolized, allowing it to be used as an indirect marker of intestinal function (Scharrer and Grenacher, 2000). D-xylose (0.5 g/kg BW) was fed to each calf by adding it to the morning feeding of WM (all milk was consumed). The dose of D-xylose provided was based on the weight of the calf taken between 1830 and 1900 h on d 4. On d 5, blood samples were taken before (0 h) and 2, 4, 6, 8, and 12 h after feeding. Calves were fasted for the entire blood sampling time but were allowed access to free-choice water. Blood samples were collected via jugular venipuncture into a 10-mL tube with EDTA (Monoject, Covidien, Mansfield, MA) and centrifuged at $3,300 \times g$ at 4°C for 20 min. Plasma was aspirated into 2 1-mL aliquots and stored at -20°C until further analysis to measure D-xylose (as described by Merritt and Duely,

1983) and glucose concentrations (Wako Autokit Glucose Assay, Wako Diagnostics, Mountain View, CA).

Statistical Analysis

Calf performance was statistically analyzed based on the treatments of their dams. Data were analyzed using the MIXED procedure of SAS 9.4 (SAS Institute, Inc., Cary, NC). Normality was tested using the Shapiro–Wilk test. If the data did not fit a normal distribution and could not be transformed to fit a normal distribution, the GLIMMIX procedure was used. The GLIMMIX procedure provides a better-fitting model for data that does not fit a normal distribution (SAS Institute, Inc., Cary, NC). Degrees of freedom were calculated using the Kenward–Roger approximation option for the MIXED and GLIMMIX procedures. Covariates used were dam parity and calf sex (1,2). If $P > 0.25$ for a covariate measurement, it was removed from the statistical model. For repeated measures, the model structure was selected based on which covariance matrix produced the lowest Bayesian information criterion. Covariance structures measured were variance components, autoregressive –1, compound symmetry, Toeplitz, and unstructured. Significance was declared at $P < 0.05$ and trends at $0.05 \leq P \leq 0.10$. Outliers within the data were declared at ± 3.0 SD from the mean and subsequently removed from the dataset. Orthogonal contrasts were used to determine the effect of the level of DCAD, the effect of NA, or any interaction effects. For calf data, initial BW was not used as a covariate because of its potential to be an effect of treatment. Initial and final calf BW, 0-h and 24-h IgG concentrations and AEA, initial and final ketones, and initial and final glucose were analyzed according to the following model:

$$Y_{ij} = \mu + B_i + \text{TRT}_j + \beta X_{ij} + e_{ij},$$

where Y_{ij} = the dependent variable, μ = the overall mean, B_i = the random effect of the i th block ($i = 1 \dots 10$), TRT_j = the fixed effect of the j th treatment level of the dam ($j = 1 \dots 4$), β = the regression (covariate coefficient), X_{ij} = the covariate measurement, and e_{ij} = the residual error term. In this model, the random effect of calf within the block subclass was used as the error term for the effect of treatment.

Calf intakes, ADG, FE, weekly BW, weekly skeletal measurements (hip height, withers height, heart girth, and body length), weekly ketones, and weekly glucose were analyzed as repeated measures according to the following model:

$$Y_{ijkl} = \mu + B_i + \text{TRT}_j + W_k + \beta X_{ij} + \text{TRT} \times W_{jk} + e_{ijkl},$$

where Y_{ijkl} = the dependent variable, μ = the overall mean, B_i = the random effect of the i th block ($i = 1 \dots 10$), TRT_j = the fixed effect of the j th treatment level ($j = 1 \dots 4$), W_k = the fixed effect of the k th week relative to calving ($k = 1 \dots 5$), β = the regression (covariate coefficient), X_{ij} = the covariate measurement, $TRT \times W_{jk}$ = the fixed interaction between the j th treatment level and the k th week, and e_{ijkl} = the residual error term.

Blood glucose and xylose concentrations on d 5 of life were analyzed as repeated measures according to the following model:

$$Y_{ijkl} = \mu + B_i + TRT_j + H_k + \beta X_{ij} + TRT \times H_{jk} + e_{ijkl},$$

where Y_{ijkl} = the dependent variable, μ = the overall mean, B_i = the random effect of the i th block ($i = 1 \dots 10$), TRT_j = the fixed effect of the j th treatment level of the dam ($j = 1 \dots 4$), H_k = the fixed effect of the k th hour relative to xylose feeding on d 5 of life ($k = 0, 2, 4, 6, 8, 12$), β = the regression (covariate coefficient), X_{ij} = the covariate measurement, $TRT \times H_{jk}$ = the fixed interaction between the j th treatment level and the k th hour, and e_{ijkl} = the residual error. In this model, the random effect of calf within the block subclass was used as the error term for the effect of treatment. Initial (0 h) blood glucose or xylose was used as covariates when analyzing their respective hourly values.

RESULTS AND DISCUSSION

The objective of this trial was to investigate the effects of 2 levels of DCAD and NA supplementation on the cow and their subsequent effects on the calf after they consumed their dam's colostrum. Details regarding the effects of these treatments on the cows, as well as their colostrum production and composition, are in a companion paper (Stahl et al., 2026). We expected that this experiment would help determine, in multiparous Jersey cows: (1) a level of DCAD that would improve colostrum quantity, and (2) the effects of NA supplementation, aiming to improve colostrum quality. In calves, we hypothesized that consuming colostrum from dams supplemented with NA would have improved FE based on the results of Aragona et al. (2020). There are few studies evaluating prepartum DCAD diets and their effects on calf health and performance in Holsteins (Collazos et al., 2017; Rajaeerad et al., 2020; Melendez et al., 2021; Zimpel et al., 2021), but to our knowledge, no one has investigated prepartum DCAD levels for Jersey cows with respect to calves. Also, there have been studies evaluating the effects of NA in various stages of lactation (Erickson et al., 1992; Aschemann et al., 2012; Standish et al., 2024), but few have looked at the effects on colostrum and subsequent calf performance (Aragona et al., 2020). Also,

there are few studies evaluating prepartum DCAD diets and their effects on calf health and performance, but to our knowledge, no one has investigated different DCAD levels for prepartum Jersey cows.

Length of Time on Treatment, Number of Calves Enrolled, and Calf Health

In total, 5 bull calves and 4 heifer calves were born from cows on $-40/NA^-$, 7 bull calves and 3 heifer calves were born from cows on $-40/NA^+$, 4 bull calves and 6 heifer calves were born from cows on $-80/NA^-$, and 6 bull calves and 4 heifer calves were born from cows on $-80/NA^+$. One set of heifer twins was born from a cow on $-40/NA^-$. One stillborn heifer was born from a cow on $-80/NA^-$. The remaining 9 calves not used on trial came from dams that produced <2 L maternal colostrum: 1 bull calf and 2 heifer calves came from cows on $-40/NA^-$, 1 bull calf and 1 heifer calf came from cows on $-40/NA^+$, 1 bull calf and 1 heifer calf came from cows on $-80/NA^-$, and 1 bull calf and 1 heifer calf came from cows on $-80/NA^+$. All enrolled calves ($n = 31$) remained on the study for 6 wk. Calves from $-40/NA^-$ cows were on the study an average of 38.5 d, calves from $-40/NA^+$ cows were on the study an average of 38.3 d, calves from $-80/NA^-$ cows were on study an average of 39 d, and calves from $-80/NA^+$ were on the study an average of 38.5 d.

Eight calves received organic medical treatment, 2 calves from cows fed $-40/NA^-$ (1 for diarrhea, and 1 for throat irritation from esophageal feeding), 1 calf from cows fed $-40/NA^+$ (1 for throat irritation from esophageal feeding), 3 calves from cows fed $-80/NA^-$ (diarrhea), and 2 calves from cows fed $-80/NA^+$ (diarrhea). Throat irritation was treated with milk of magnesia for 2 d; and diarrhea was treated with Get Well (CowMaster, LLC, Saxapahaw, NC), injectable B-complex, and vitamin C for 2 d.

Chemical Composition of Diets and Feed Ingredients

The composition and nutrient analysis of calf starter and WM are presented in Table 1. The calf starter did not change in composition, and the WM values $\pm$ SD presented include every milk sample taken when bottles were filled throughout the 18-mo trial.

Calf Intake, Growth, and Blood Metabolites

The LSM for colostrum yield, IgG concentration (g/L; Stahl et al., 2026), calf intake of colostrum, IgG, and bioactive compounds are presented in Table 2. There was a trend ($P = 0.07$) for an interaction on IgG concentration, with the greatest concentration at $-80/NA^-$ and the lowest concentration at $-80/NA^+$. There were no differ-

ences observed in colostrum yield (Stahl et al., 2026), the amount of colostrum fed to calves, or intakes of insulin, TGF β -1, and TGF β -2. We observed a lower ($P < 0.04$) IgG intake in calves that were born of cows supplemented with the -80 DCAD diets as compared with the -40 DCAD diets. Calves fed colostrum from cows fed the -80 DCAD diets consumed 115 g less IgG than calves fed colostrum from cows fed the -40 DCAD diet. There was a trend ($P = 0.09$) for an interaction for IgG intake, with calves born to cows fed $-40/\text{NA}^+$ consuming the greatest amount of IgG (411.3 g) and calves born to cows fed $-80/\text{NA}^+$ consuming 50% less IgG (200.5 g). This could be due to cows fed the $-40/\text{NA}^+$ treatment having a greater colostrum IgG concentration (Table 2) as compared with the $-80/\text{NA}^+$, which was supported by Aragona et al. (2020), who observed that Holstein cows fed 32 g/d NA for 28 d produced more IgG than cows fed either 0, 16, or 48 g/d in a diet with ~ 0 mEq/kg DCAD. However, in the current study, colostrum yield was not different across treatments. We are not aware of any studies that investigated NA supplementation and DCAD and how they affected IgG intake.

There were no differences in concentrations or yields of bioactive compounds in colostrum (Stahl et al., 2026). However, when considering amount of colostrum fed, we observed changes in specific bioactive compounds. We observed an interaction ($P < 0.02$) for IGF-1, with the greatest intake of IGF-1 in calves that came from cows fed $-40/\text{NA}^+$. For LF, we observed higher ($P < 0.02$) LF intake in calves that came from cows supplemented with NA. To our knowledge, there are no data with either prepartum DCAD diets or prepartum NA supplementation on bioactive compounds in colostrum,

nor subsequent intake of these compounds in the calf. We hypothesize that these data can explain the results seen with FE in the first 3 wk of life for Holstein calves that was observed by Aragona et al. (2020). Bioactive compounds found in mammary secretions can affect the calf gastrointestinal (GI) tract by directly affecting intestinal epithelium (via cell proliferation or differentiation), being transported across the intestinal wall into circulation, or both (Baumrucker and Bruckmaier, 2014; Ontsouka et al., 2016).

Many of these bioactive compounds are transported from the colostrum to the calf GI tract via cell surface receptors and receptor-like transporters within the intestinal epithelium (Ontsouka et al., 2016). Insulin-like growth factor-1 has been shown to increase the growth of neonatal intestinal tissue within the first week of life (Baumrucker and Blum, 1993; Pyo et al., 2020). For LF, the earliest known function was as a bacteriostat due to its ability to bind Fe, thus removing an essential nutrient for bacteria (Smith and Schanbacher, 1977; Bullen et al., 1978). Due to this, LF is involved in the innate immune system (Prgomet et al., 2007), and can possibly improve calf growth and FE through improved intestinal health. Several studies indicate that LF stimulates cell growth (Nichols et al., 1987; Azuma et al., 1989; Byatt et al., 1990; Hagiwara et al., 1995), which is thought to be achieved through the ability of LF to bind DNA and activate transcription (Fleet, 1995). These effects on cell proliferation are indicative of increased intestinal functionality, which we hypothesized is confirmed by the xylose challenge data in the current experiment.

We observed no differences in 0-h IgG, 24-h IgG, and AEA (Table 3). There is no explanation as to why 0-h

Table 2. Calf intake¹ of colostrum, IgG, and bioactive compounds from cows supplemented -40 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid or -80 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid prepartum

Item	Treatment ²				SEM	P-value		
	$-40/\text{NA}^-$	$-40/\text{NA}^+$	$-80/\text{NA}^-$	$-80/\text{NA}^+$		DCAD	NA ³	Int ⁴
Colostrum yield ⁵ (kg)	6.16	4.84	4.31	5.77	1.33	0.72	0.95	0.29
Colostrum IgG ⁵ (g/L)	103.4	117.8	125.5	83.7	15.37	0.67	0.35	0.07
Colostrum fed (L)	3.59	3.60	3.33	3.22	0.24	0.13	0.83	0.76
IgG intake (g)	331.24 ^{ab}	411.28 ^a	312.23 ^{ab}	200.50 ^b	—	0.04	0.66	0.09
IgG intake 95% CI	[268.8–549.3]	[258.5–506.4]	[201.2–407.9]	[163.2–281.0]				
Insulin intake (μg)	332.66	358.04	359.62	321.82	76.00	0.95	0.94	0.64
IGF-1 intake (μg)	1,524.7 ^b	2,362.2 ^a	2016.5 ^{ab}	1,820.7 ^{ab}	223.27	0.91	0.19	0.02
Lactoferrin intake (g)	2.53	7.22	3.45	6.51	1.37	0.72	0.02	0.48
TGF β -1 intake (μg)	45.91	38.59	47.76	42.94	4.76	0.49	0.24	0.78
TGF β -2 intake (μg)	291.66	267.16	263.34	260.81	16.83	0.26	0.46	0.47

^{a,b}Means with different superscripts differ ($P < 0.10$).

¹Intake is based on the colostrum amount fed to calves.

² $-40/\text{NA}^-$ = without NA; $-40/\text{NA}^+$ = with NA; $-80/\text{NA}^-$ = without NA; $-80/\text{NA}^+$ = with NA.

³NA = nicotinic acid.

⁴Int = interaction.

⁵Data from Stahl et al. (2026).

Table 3. Immunoglobulin G concentrations and apparent efficiency of absorption of calves born from dams supplemented -40 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid or -80 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid prepartum

Item	Treatment ^{1,2}				SEM	P-value		
	$-40/\text{NA}^-$	$-40/\text{NA}^+$	$-80/\text{NA}^-$	$-80/\text{NA}^+$		DCAD	NA ³	Int ⁴
0 h IgG (g/L)	0.12	1.71	0.17	0.21	0.80	0.35	0.29	0.31
24 h IgG (g/L)	38.08	42.30	35.83	39.79	0.55	0.69	0.55	0.99
AEA ⁵ (%)	33.37	34.01	32.55	38.43	4.96	0.70	0.53	0.57

¹Treatments fed to dams prepartum: $-40/\text{NA}^-$ = without NA; $-40/\text{NA}^+$ = with NA; $-80/\text{NA}^-$ = without NA; $-80/\text{NA}^+$ = with NA.

²Number of calves per treatment: 8 from $-40/\text{NA}^-$, 8 from $-40/\text{NA}^+$, 7 from $-80/\text{NA}^-$, and 8 from $-80/\text{NA}^+$.

³NA = nicotinic acid.

⁴Int = interaction.

⁵Apparent efficiency of absorption (AEA): $\{[24 \text{ h plasma IgG (g/L)} \times \text{BW (kg)} \times 0.09]/\text{IgG intake (g/L)}\} \times 100$ (Quigley et al., 1998).

IgG levels were higher in $-40/\text{NA}^+$ calves compared with the other treatments. Generally, observing a 0-h IgG concentration in calves could be due to transplacental transfer of IgG or fetal production (Chigerwe et al., 2008) and has been reported in other studies (Wilm et al., 2018; Aragona et al., 2020). Although calves consumed more IgG from colostrum produced by cows fed the -40 DCAD diets compared with the -80 DCAD diets (371.3 g, 256.4 g, respectively), we observed no difference in IgG concentration at 24 h and AEA, which is supported by Morrill et al. (2010), Collazos et al. (2017), and Zimpel et al. (2021) for DCAD and Aragona et al. (2016, 2020) for NA supplementation.

The LSM for the d 5 xylose challenge performance of calves are presented in Table 3. We observed a significant ($P = 0.02$) effect of prepartum NA supplementation resulting in increased xylose absorption in their calves (Table 4) and a treatment-by-hour effect ($P = 0.03$; Figure 1). The effect of hour was different ($P < 0.0001$), and the overall effect of treatment was also different ($P < 0.05$; data not shown). Because of the highly significant effect of hour, the treatment-by-hour interaction was different ($P = 0.03$). When comparing data within each hour, no

differences were observed. Regardless of treatment, xylose peaked in all calves at 4 h postdose. We observed a tendency ($P = 0.10$) to decrease circulating glucose during the xylose challenge from calves that were born to cows supplemented with -40 DCAD. It is not clear why these results were observed. We hypothesized that this increased intestinal absorption (as measured by plasma xylose concentrations) was a result of select bioactive compounds that enhance intestinal epithelial cell development. In the current study, this was due to LF in colostrum from NA-supplemented cows and IGF-1 from cows fed $-40/\text{NA}^+$ as compared with IGF-1 in colostrum from cows fed $-40/\text{NA}^-$.

Calf feed intake and BW measurements are presented in Table 5. We observed no differences in starter intake, milk solids intake, free water intake, total DMI, weekly BW, final BW, all skeletal measurements, weekly glucose, final glucose, and weekly ketones. Milk solids intake was 540 g/d, which is considered a moderate feeding rate (NASEM, 2021). Expressed differently, the minimum amount of milk solids recommended to be fed is 1.5% of birth BW, and in the current study, we fed 1.88%, 1.88%, 1.95%, and 1.88% of birth BW as milk solids for $-40/$

Table 4. Xylose challenge performance on d 5 of life of calves born from dams supplemented -40 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid or -80 mEq/kg DM DCAD diet ± 23 g/d nicotinic acid prepartum

Item	Treatment ^{1,2}				SEM	P-value			
	$-40/\text{NA}^-$	$-40/\text{NA}^+$	$-80/\text{NA}^-$	$-80/\text{NA}^+$		DCAD	NA ³	Int ⁴	Trt $\times$ h ⁵
0 h xylose (mmol/L)	1.08	1.13	1.07	1.20	0.07	0.62	0.22	0.52	—
Xylose (mmol/L·h)	2.35	2.60	2.49	2.61	0.08	0.24	0.02	0.31	0.03
0 h glucose (mg/dL)	93.8	95.1	93.6	92.0	5.08	0.71	0.98	0.74	—
Glucose (mg/dL·h)	100.3	100.1	106.0	102.3	2.43	0.10	0.36	0.49	0.002

¹Treatments fed to dams prepartum: $-40/\text{NA}^-$ = without NA; $-40/\text{NA}^+$ = with NA; $-80/\text{NA}^-$ = without NA; $-80/\text{NA}^+$ = with NA.

²Number of calves per treatment: 8 from $-40/\text{NA}^-$, 8 from $-40/\text{NA}^+$, 7 from $-80/\text{NA}^-$, and 8 from $-80/\text{NA}^+$.

³NA = nicotinic acid.

⁴Int = interaction.

⁵Trt $\times$ h = treatment-by-hour interaction.

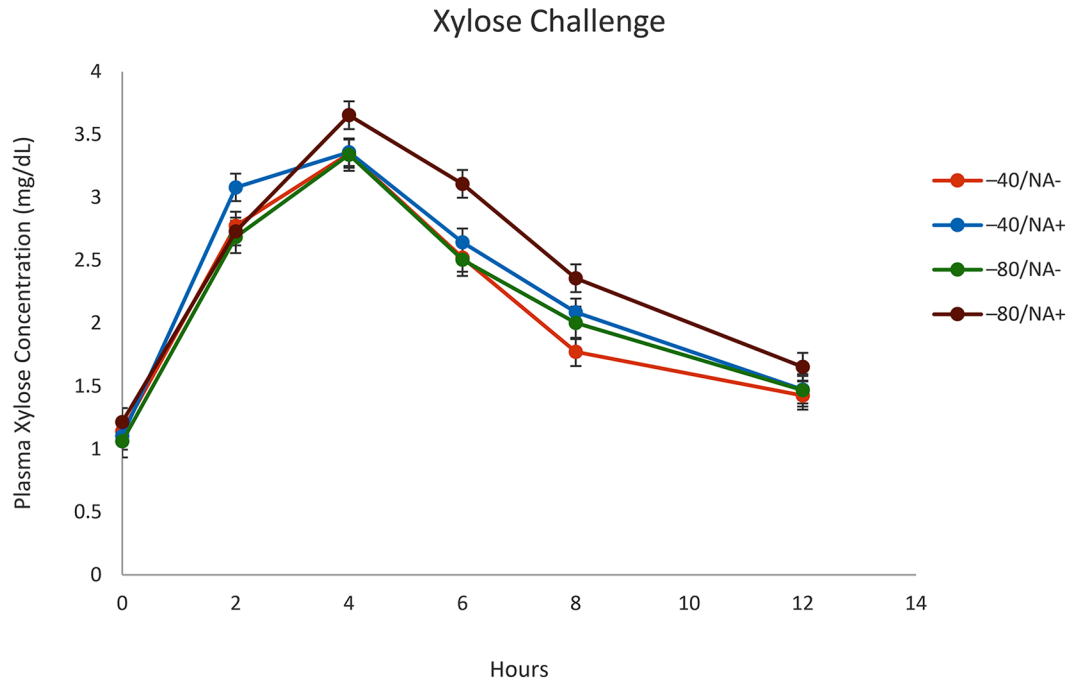


Figure 1. Average xylose concentration for calves born from dams supplemented prepartum with a -40 mEq/kg DM DCAD diet $\pm$ 23 g/d nicotinic acid ($-40/\text{NA}^-$ and $-40/\text{NA}^+$) or -80 mEq/kg DM DCAD diet $\pm$ 23 g/d nicotinic acid ($-80/\text{NA}^-$ and $-80/\text{NA}^+$).

NA^- , $-40/\text{NA}^+$, $-80/\text{NA}^-$, and $-80/\text{NA}^+$, respectively. The results observed for starter intake, milk solids intake, and free water intake are supported by Aragona et al. (2020) for NA supplementation. The results observed for starter intake could partially be supported for DCAD

by Zimpel et al. (2021), although these researchers only measured starter intake in calves from d 21 to d 56 of life. To our knowledge, no studies have investigated the effect of prepartum DCAD concentrations fed to cows on the subsequent milk solids and water intake of their

Table 5. Intake and performance¹ of calves born from dams supplemented -40 mEq/kg DM DCAD diet $\pm$ 23 g/d nicotinic acid or -80 mEq/kg DM DCAD diet $\pm$ 23 g/d nicotinic acid prepartum

Item	Treatment ^{2,3}				SEM	P-value			
	$-40/\text{NA}^-$	$-40/\text{NA}^+$	$-80/\text{NA}^-$	$-80/\text{NA}^+$		DCAD	NA^4	Int ⁵	Trt $\times$ wk ⁶
Starter intake (g/d)	103.60	144.44	146.20	125.40	30.0	0.70	0.75	0.32	0.31
Milk solids intake (g/d)	541.0	540.0	540.0	540.0	0.50	0.69	0.30	0.43	0.61
Free water intake (kg/d)	1.22	0.87	1.42	1.03	0.34	0.57	0.26	0.95	0.25
Total DMI (kg/d)	0.65	0.67	0.68	0.67	0.03	0.56	0.92	0.54	0.41
Birth BW (kg)	28.73	28.75	27.72	28.73	1.07	0.60	0.65	0.60	—
BW (kg)	38.01	37.10	37.57	37.38	1.01	0.93	0.59	0.71	0.29
Final BW (kg)	45.15	45.22	45.38	44.09	1.98	0.80	0.77	0.70	—
ADG (kg/d)	0.52	0.48	0.49	0.37	0.04	0.05	0.06	0.34	0.001
ADG/DMI	0.74 ^{a,x}	0.73 ^{a,x,y}	0.67 ^{c,y}	0.58 ^{b,d}	0.03	<0.0001	0.02	0.03	0.003

^{a,b}Means with different superscripts vary ($P < 0.0001$).

^{c,d}Means with different superscripts vary ($P < 0.02$).

^{x,y}Means with different superscripts vary ($P < 0.10$).

¹Weekly blood and BW sampling schedule: calves born after 1630 h on Thursday until Monday at 0700 h were sampled on Tuesday at 0700 h; calves born after 0700 h on Monday until Thursday at 1630 h were sampled on Friday at 1630 h.

²Treatments fed to dams prepartum: $-40/\text{NA}^-$ = without NA; $-40/\text{NA}^+$ = with NA; $-80/\text{NA}^-$ = without NA; $-80/\text{NA}^+$ = with NA.

³Number of calves per treatment: 8 from $-40/\text{NA}^-$, 8 from $-40/\text{NA}^+$, 7 from $-80/\text{NA}^-$, and 8 from $-80/\text{NA}^+$.

⁴NA = nicotinic acid.

⁵Int = interaction.

⁶Trt $\times$ wk = treatment-by-week interaction.

respective calves. The results for total DMI are supported by Rajaerad et al. (2020) for DCAD and partially supported by Zimpel et al. (2021), though these researchers only reported residual DMI because they did not collect intake from d 0 to 21 of life.

For weekly and final BW, our results are supported by Weich et al. (2013) for DCAD and Aragona et al. (2016, 2020) for NA supplementation. For weekly BW, this is partially supported by Zimpel et al. (2021), though these researchers only measured BW at birth and on d 21, 42, and 56 of age. In the current experiment, we observed that -80 DCAD diets provided to prepartum cows resulted in a tendency ($P = 0.05$) to decrease calf ADG, and that NA supplementation to prepartum cows resulted in a tendency ($P = 0.06$) to decrease calf ADG. Calves in this study gained between 370 and 520 g/d. Compared with Jersey calves fed at a high plane of nutrition (28% CP, 25% fat, 568 g of solids; Ballou, 2012), calves in the present study gained 52 g/d more and consumed less milk solids. Calves were removed from the study at d 42, so if BW is projected out to d 56 using initial BW and ADG, calves weighed 201%, 193%, 199%, and 172% of initial BW. Reasons for both these decreases in ADG are not clear. However, for DCAD, it could be similar to the lag in early body growth in calves observed by Collazos et al. (2017) when cows were supplemented with -180 mEq/kg of DM for 42 d prepartum and could potentially be related to fetal programming. It is worth noting that Stahl et al. (2026) observed no differences in the main effect of DCAD on urine pH (5.91 vs. 5.87 for -40 DCAD and -80 DCAD, respectively) or DMI (10.8 vs. 10.8 kg/d for -40 DCAD and -80 DCAD, respectively), suggesting that there are other physiological changes occurring in these cows that may contribute to calf performance.

The results of the current study for FE indicate that prepartum cows supplemented with an -80 DCAD diet resulted in a decrease ($P < 0.0001$) in calf FE when compared with -40 DCAD, and that NA supplementation to prepartum cows resulted in a decrease ($P = 0.02$) in calf FE. However, there was an interaction ($P = 0.03$), and upon applying the Tukey mean separation procedure, it was found that results were similar regardless if NA was added to the -40 treatment. In the current experiment, it is important to note that we observed no differences in calf DMI, therefore, the FE results are primarily due to a tendency for both main effects to reduce ADG. Reasons for these decreases in FE are not clear, especially considering the results with xylose concentration, LF, and IGF-1 fed to calves. Also, the FE results in the current study are contradicted by previous data with NA supplementation, because Aragona et al. (2020) found increased calf FE when cows were supplemented with 32 g/d of nicotinic

acid. These researchers also saw a treatment-by-week effect that showed calves born from cows supplemented with 32 g/d had greater FE during wk 2 and 3 of life that was not observed from wk 4 to 6, hypothesizing greater intestinal development occurring in these calves in utero or through colostrum consumption. Based on this previous data, along with our results for xylose concentration, LF, and IGF-1 fed to calves, we hypothesized that we would see an increase in FE from calves that came from NA-supplemented cows. However, we observed the opposite because there was an interaction effect of the lowest FE in calves that came from cows fed $-80/NA^+$. We hypothesize that there may be a negative in utero effect with the -80 DCAD based on the lower IgG intake, ADG, and calf FE, which could negate any positive effects seen in intestinal development (xylose challenge response).

We observed no effect of treatments on skeletal measurements, skeletal gains, blood glucose, or ketones (Table 6). The results observed with skeletal measurements were partially supported for DCAD by Collazos et al. (2017), who found that calf skeletal growth in the first 62 d of life (based on hip height) was not affected by the level of DCAD provided to cows. In the current study, prepartum NA supplementation did not affect calf skeletal measurements and gains, which was contradicted by Aragona et al. (2020), who observed that NA supplementation to prepartum cows increased both skeletal measurements and weekly gains in their respective calves. No differences were observed for glucose concentrations in the current experiment, which is supported for DCAD by Collazos et al. (2017), and to our knowledge, there are no data to support or contradict these results for NA. The results for final ketones in the current trial are partially supported by Collazos et al. (2017), who saw lower ketone concentrations in calves born to dams supplemented with a -180 mEq/kg DM DCAD. As for NA, the results for ketones were contradicted by Aragona et al. (2020), as they observed calf ketone concentration increasing quadratically when cows went from 0 g/d to 32 g/d NA and decreasing from 32 g/d to 48 g/d. In the current study, we observed an interaction effect for final ketones, where the greatest final ketone concentrations in calves were from $-40/NA^+$ and $-80/NA^-$ treatments. Increased ketone concentration suggests that the calf is becoming more reliant on gluconeogenesis and can be used as an indicator of rumen development. Quigley et al. (1991) observed greater ketone and reduced glucose concentrations as calves adapt to consuming solid feed.

Table 6. Skeletal measurements, blood glucose, and ketones¹ of calves born from dams supplemented with –40 mEq/kg DM DCAD diet +/- 23 g/d nicotinic acid or –80 mEq/kg DM DCAD diet +/- 23 g/d nicotinic acid prepartum

Item	Treatment ^{2,3}				SEM	P-value			
	–40/NA [–]	–40/NA ⁺	–80/NA [–]	–80/NA ⁺		DCAD	NA ⁴	Int ⁵	Trt × wk ⁶
Initial heart girth (cm)	70.56	69.45	69.28	71.20	0.96	0.79	0.70	0.09	—
Heart girth (cm)	76.76	75.48	75.58	76.32	1.00	0.85	0.80	0.26	0.07
Heart girth daily gain (cm/d)	0.32	0.33	0.34	0.28	0.02	0.44	0.40	0.13	0.08
Final heart girth (cm)	82.61	81.01	81.38	80.62	1.30	0.50	0.40	0.72	—
Initial withers height (cm)	72.48	71.92	71.41	72.21	0.86	0.62	0.89	0.38	—
Withers height (cm)	76.12	75.44	74.99	75.98	0.63	0.60	0.82	0.14	0.95
Withers height daily gain (cm/d)	0.18	0.19	0.20	0.21	0.02	0.37	0.82	0.88	0.55
Final withers height (cm)	79.48	78.70	78.39	79.39	0.60	0.70	0.86	0.11	—
Initial hip height (cm)	74.66	73.60	73.51	74.95	0.88	0.90	0.84	0.13	—
Hip height (cm)	78.21	77.37	77.77	78.38	0.65	0.62	0.87	0.20	0.38
Hip height daily gain (cm/d)	0.20	0.21	0.22	0.19	0.02	0.96	0.65	0.23	0.20
Final hip height (cm)	82.15	81.14	81.20	81.75	0.68	0.79	0.75	0.22	—
Initial body length (cm)	58.21	57.53	56.53	58.31	1.19	0.68	0.66	0.25	—
Body length (cm)	64.03	63.49	63.14	64.55	0.99	0.92	0.67	0.27	0.51
Body length daily gain (cm/d)	0.28	0.28	0.37	0.28	0.04	0.28	0.33	0.27	0.09
Final body length (cm)	69.68	68.70	69.01	69.74	1.13	0.87	0.93	0.43	—
Initial glucose (mg/dL)	93.76	95.11	93.58	91.99	5.08	0.71	0.98	0.74	—
Glucose (mg/dL)	104.41	99.38	100.21	99.35	2.26	0.32	0.23	0.33	0.09
Final glucose (mg/dL)	103.05	98.43	96.81	102.10	5.09	0.79	0.95	0.30	—
Initial ketones (mmol/L)	0.30	0.32	0.30	0.30	0.03	0.70	0.73	0.84	—
Ketones (mmol/L)	0.35	0.36	0.35	0.35	0.01	0.54	0.79	0.23	0.32
Final ketones (mmol/L)	0.37	0.41	0.43	0.38	0.02	0.53	0.93	0.03	—

¹Weekly blood and skeletal measurement sampling schedule: calves born after 1630 h on Thursday until Monday at 0700 h were sampled on Tuesday at 0700 h; calves born after 0700 h on Monday until Thursday at 1630 h were sampled on Friday at 1630 h.

²Treatments fed to dams prepartum: –40/NA[–] = without NA; –40/NA⁺ = with NA; –80/NA[–] = without NA; –80/NA⁺ = with NA.

³Number of calves per treatment: 8 from –40/NA[–], 8 from –40/NA⁺, 7 from –80/NA[–], and 8 from –80/NA⁺.

⁴NA = nicotinic acid.

⁵Int = interaction.

⁶Trt × wk = treatment-by-week interaction.

CONCLUSIONS

Bioactive compounds consumed via colostrum that aid in enhancing the intestinal development of calves were increased when NA was fed to prepartum cows. Immunoglobulin G intake was twice the amount for calves born of cows on the –40/NA⁺ treatment compared with calves born of cows fed the –80/NA⁺ treatment. Whereas IgG intake differed due to the level of DCAD, calf 24-h plasma IgG concentration was not different. The observed increases in the intake of colostrum IGF-1 (–40/NA⁺ cows) and LF (NA-supplemented cows) likely supported the development of the intestinal epithelium, which was assessed by calf plasma xylose concentrations. The –80 DCAD diets fed to cows tended to result in lesser ADG in calves fed their colostrum as compared with calves fed colostrum from cows fed the –40 DCAD diets. Nicotinic acid supplementation of cows tended to result in lesser ADG in calves fed their colostrum. There was an interaction for improved FE in calves fed colostrum from cows fed the –40/NA[–] diet. Calves consuming colostrum from cows on the –40 DCAD diets resulted in improved FE, whereas calves fed colostrum from NA-fed cows had reduced FE. Future work needs to be done to

further investigate levels of DCAD that will result in the best response for Jersey cows. With NA supplementation, many results in this experiment indicated that it was beneficial in the diets of prepartum Jersey cows. Future work should be done in this area with this breed, investigating incremental dosing to determine a dose that is optimal for prepartum, multiparous Jersey cows and the subsequent effects on their calves.

NOTES

This experiment was reviewed and approved by the University of New Hampshire Institutional Animal Care and Use Committee (Protocol #200202). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AEA = apparent efficiency of absorption; FE = feed efficiency; GI = gastrointestinal; NA = nicotinic acid; WM = whole milk.

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